Resistance and resilience

Ecosystems are often strong enough to withstand some natural external disturbances and retain their equilibrium. Some are more resistant to disturbance than others, and have adapted to the particular disturbances normally associated with their environment. In some forest ecosystems, for example, the periodic fires caused by electrical storms cause only a minor imbalance in the ecosystem.

Even when severely disrupted by external disturbances, some ecosystems have a resilience that enables them to recover. However, other ecosystems are more fragile, and if they are disturbed may never be able to regain equilibrium. The resistance and resilience of an ecosystem is generally thought to be related to its biodiversity. If, for example, there is only one species of plant performing a particular function in the system, and that species is not frost-resistant, an abnormally severe winter could deplete the species enough to have a major impact on the system as a whole. In contrast, if there are several species with that role in the system, it is more likely that one will be resistant to the disturbance.

The human factor

Some disturbances can be severe enough to be catastrophic for an ecosystem, damaging it beyond the point of recovery and so causing a permanent change in its identity, or even its demise. The fear is that much of the disturbance caused by human activity has the potential to cause such permanent damage, particularly when it involves the wholesale destruction of habitat and the consequent depletion of its biodiversity. In addition, some have suggested that human influence has created a new category of ecological systems, dubbed “techno-ecosystems.” For example, “cooling ponds” are human-made ponds, built to cool down nuclear power plants, but they have become ecosystems for aquatic organisms.